Feature Extraction Power Analysis of VGG16 Before and After Transfer Learning

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1 Introduction

Convolutional Neural Networks (CNNs) pretrained on large datasets such as ImageNet learn powerful and generic feature representations. This experiment analyzes the feature extraction capability of VGG16 before and after transfer learning using the MNIST digit dataset.

High-dimensional feature vectors are projected to a 2D plane using:

- Principal Component Analysis (PCA)
- t-distributed Stochastic Neighbor Embedding (t-SNE)

2 Methodology

2.1 Step 1: Pretrained Feature Extraction

The convolutional base of VGG16 pretrained on ImageNet was used without modification. MNIST images were resized to 224×224 and converted to RGB format before feature extraction.

2.2 Step 2: Transfer Learning

VGG16 was fine-tuned on the MNIST dataset. Only the final classification layers were retrained.

2.3 Step 3: Dimensionality Reduction

The extracted high-dimensional feature vectors were reduced to 2D using:

- PCA: Linear dimensionality reduction technique
- t-SNE: Non-linear manifold learning method

3 Results

3.1 Before Transfer Learning

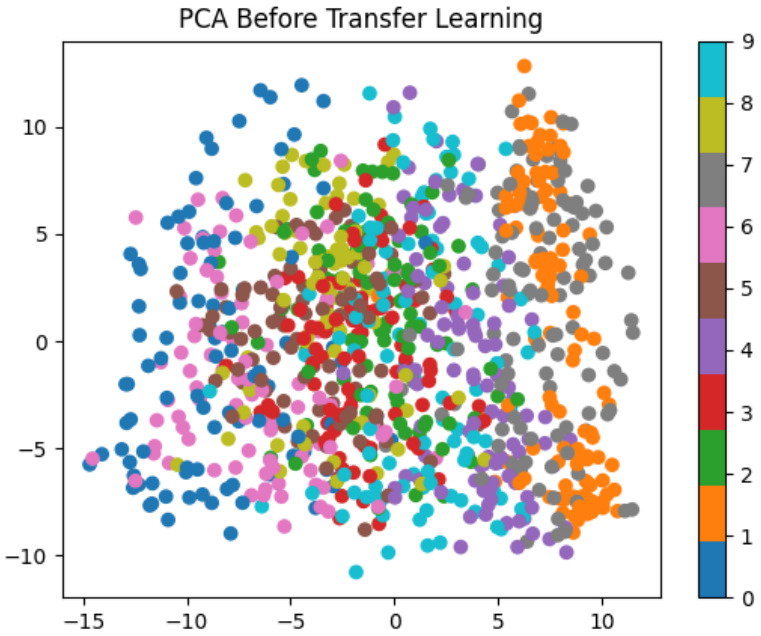

Figure 1: PCA Projection Before Transfer Learning

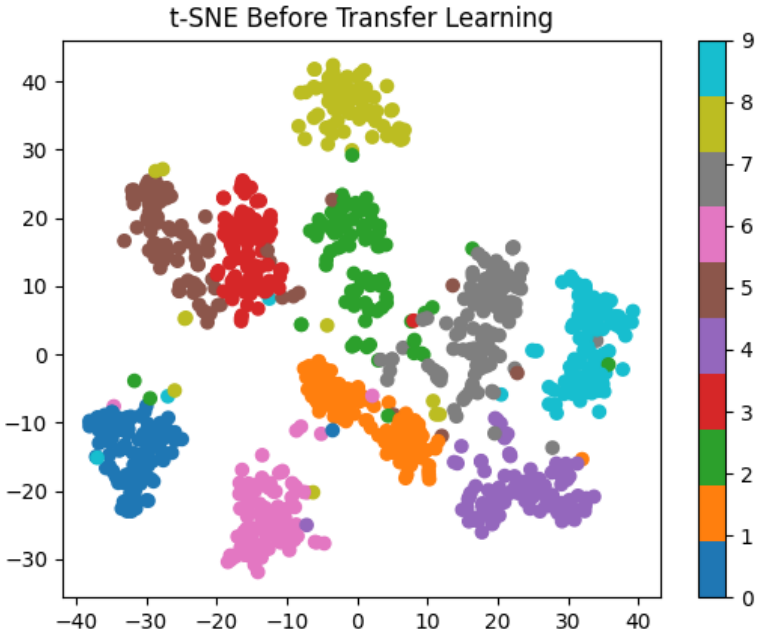

Figure 2: t-SNE Projection Before Transfer Learning

Observation: Digits are partially separable but clusters overlap significantly. The pre-trained model captures general edge and texture features, but not digit-specific representations.

3.2 After Transfer Learning

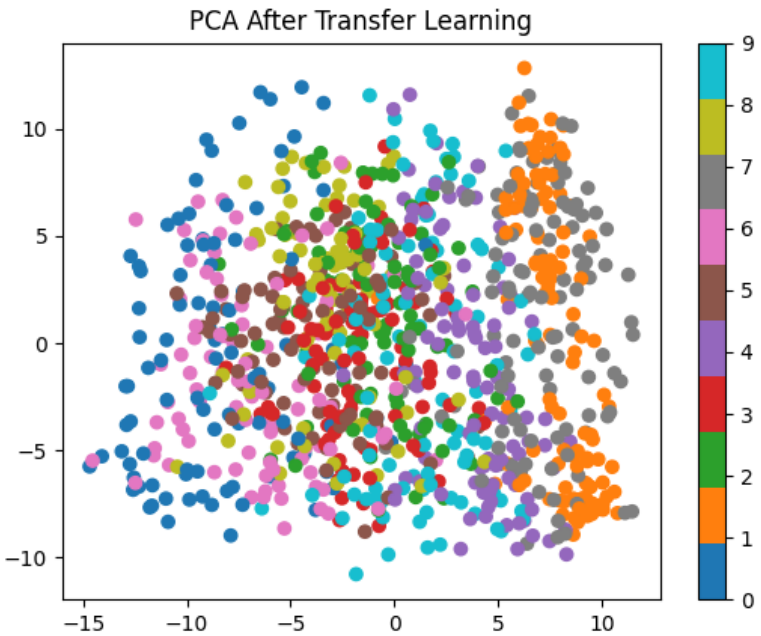

Figure 3: PCA Projection After Transfer Learning

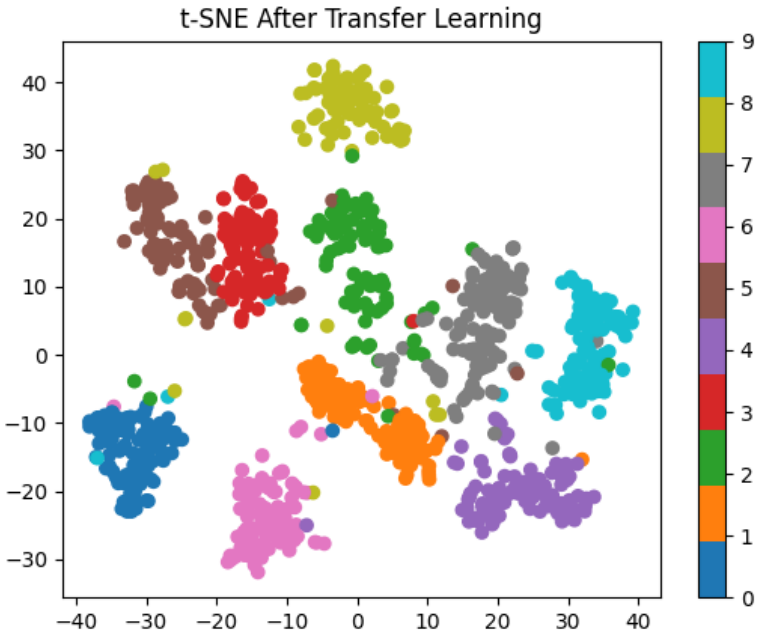

Figure 4: t-SNE Projection After Transfer Learning

Observation: Digit clusters become clearly separable. Feature vectors are more compact and class-discriminative.

4 Discussion

4.1 Feature Extraction Power Before Transfer Learning

- Extracted generic visual features
- Moderate clustering
- Overlapping class boundaries

4.2 Feature Extraction Power After Transfer Learning

- Highly discriminative digit features
- Clear cluster separation
- Improved class compactness

4.3 PCA vs t-SNE

- PCA preserves global variance structure
- t-SNE reveals local cluster structure
- t-SNE shows clearer class separation

5 Conclusion

Transfer learning significantly improves the feature extraction capability of VGG16 for MNIST digit classification. The 2D projections confirm that fine-tuned features are more class-discriminative compared to the original ImageNet pretrained features.

6 Appendix: Python Implementation

```
from sklearn.decomposition import PCA
from sklearn.manifold import TSNE

# features = extracted feature vectors
pca = PCA(n_components=2)
pca_result = pca.fit_transform(features)

tsne = TSNE(n_components=2, random_state=42)
tsne_result = tsne.fit_transform(features)
```